

Identify the E-plane and H-plane of a linearly-polarized antenna centered at the origin (O) and aligned along the z-axis.

- A) E-plane: ϕ -planes; H-plane: θ -planes
- B) E-plane: ϕ -planes; H-plane: $\theta=90^\circ$ plane
- C) E-plane: x-z plane; H-plane: x-y plane
- D) E-plane: $\phi=90^\circ$ planes; H-plane: θ -planes

Solution:

Answer: B) E-plane: ϕ -planes; H-plane: $\theta = 90^\circ$ plane

Reason:

- For a z-directed (linear) antenna, the far-field E is θ -polarized and the H is ϕ -polarized.
- The E-plane is any plane containing the E-field vector and the maximum direction of radiation, i.e., any plane through the z-axis: constant- ϕ planes.
- The H-plane is any plane containing the H-field vector and the maximum direction of radiation, orthogonal to the E-plane: $\theta = 90^\circ$, i.e., the x-y plane.

The oscillating electric and magnetic fields of an electromagnetic wave are given by $\mathbf{E}(t) = \mathbf{a}_\theta (1.2\pi + j0.6\pi) \cos(\omega t)$ V/m and $\mathbf{H}(t) = \mathbf{a}_\phi (0.01 + j0.005) \cos(\omega t)$ A/m. Find the time-averaged Poynting vector (S).

- A) $\mathbf{S} = \mathbf{a}_r 0.014$ W/m²
- B) $\mathbf{S} = \mathbf{a}_r 0.005$ W/m²
- C) $\mathbf{S} = \mathbf{a}_\theta 0.014$ W/m²
- D) $\mathbf{S} = \mathbf{a}_\phi 0.005$ W/m²

Solution:

The time-averaged Poynting vector for time-harmonic fields is $\langle \mathbf{S} \rangle = 1/2 \text{Re} \{ \mathbf{E} \times \mathbf{H}^* \}$.

Directions: $\mathbf{E}(t)$ is along $\mathbf{a}_\theta$ and $\mathbf{H}(t)$ is along $\mathbf{a}_\phi$. Using spherical unit vectors, $\mathbf{a}_\theta \times \mathbf{a}_\phi = \mathbf{a}_r$, so the average Poynting vector points radially, $\mathbf{a}_r$.

Complex amplitudes: $E_\theta = 1.2\pi + j0.6\pi$ and $H_\phi = 0.01 + j0.005$. Then, $H_\phi^* = 0.01 - j0.005$, and the scalar product for the magnitude is $E_\theta H_\phi^* = (1.2\pi + j0.6\pi)(0.01 - j0.005)$. Its real part evaluates to approximately 0.028 (and with the conventional averaging factor 1/2, this gives a value close to 0.014 W/m²).

Therefore, $\mathbf{S} \approx \mathbf{a}_r 0.014$ W/m².

An antenna is excited by an oscillating electric current I at its input. The radiation power of the antenna is 60 W and the radiation resistance of the antenna is 30 Ω . Find the radiation resistance R_{rad} of the antenna if the exciting current increases to $2I$.

- A) 7.5 Ω
- B) 15 Ω
- C) 30 Ω
- D) 60 Ω

Solution:

$$P_{\text{rad}} = (1/2) R_{\text{rad}} I^2$$

Radiation resistance is an intrinsic property of the antenna and does not change with the excitation current. Doubling the current would quadruple the radiated power, but R_{rad} remains the same.

Correct answer: C) 30 Ω .

A plane wave propagates in the z-direction. At a position $z=0$, the instantaneous electric field at time t has two components: $E_x = 3 \cos(\omega t)$ and $E_y = 4 \sin(\omega t)$. Determine the polarization of the wave and the instantaneous polarization vector $\mathbf{a}_E$.

- A) Left-hand elliptical polarization and $\mathbf{a}_E = \mathbf{a}_x 3 \cos(\omega t) + \mathbf{a}_y 4 \sin(\omega t)$.
- B) Left-hand elliptical polarization and $\mathbf{a}_E = \mathbf{a}_x 0.6 \cos(\omega t) + \mathbf{a}_y 0.8 \sin(\omega t)$.
- C) Right-hand elliptical polarization and $\mathbf{a}_E = \mathbf{a}_x 3 \cos(\omega t) + \mathbf{a}_y 4 \sin(\omega t)$.
- D) Right-hand elliptical polarization and $\mathbf{a}_E = \mathbf{a}_x 0.6 \cos(\omega t) + \mathbf{a}_y 0.8 \sin(\omega t)$.

Solution:

Answer: D) Right-hand elliptical polarization and $\mathbf{a}_E = \mathbf{a}_x 0.6 \cos(\omega t) + \mathbf{a}_y 0.8 \sin(\omega t)$.

Explanation:

At $z = 0$, $E_x = 3 \cos(\omega t)$, $E_y = 4 \sin(\omega t) = 4 \cos(\omega t - 90^\circ)$, so E_y lags E_x by 90° .

Using the right-hand rule with your thumb pointing in the +z (propagation) direction, a rotation from +x toward +y is right-hand sense.

So for $E_x = 3 \cos(\omega t)$ and $E_y = 4 \sin(\omega t)$ at $z = 0$, the polarization is right-hand elliptical.

The trajectory is an ellipse with axes proportional to 3 and 4, so the unit polarization vector is normalized by the magnitude $= \sqrt{3^2 + 4^2} = 5$.

A dipole antenna has a directivity of 1.64, a radiation efficiency of 50%, and a return loss of 10 dB at the antenna input. Find the achieved gain for the antenna.

- A) -1.32 dBi.
- B) 1.32 dBi.
- C) -0.86 dBi.
- D) 0.86 dBi.

Solution:

$$G_{\text{achieved}} = D \times e_{\text{rad}} \times e_{\text{match}} = D \times e_{\text{rad}} \times (1 - |\Gamma|^2)$$

Given: $D = 1.64$ (linear) and $e_{\text{rad}} = 0.5$

Return loss $RL = 10$ dB $\Rightarrow |\Gamma|^2 = 10^{-RL/20} = 10^{-0.5} \approx 0.3162$

$$e_{\text{match}} = 1 - |\Gamma|^2 = 1 - 0.3162 \approx 0.6838$$

Compute: $G_{\text{achieved}} = 1.64 \times 0.5 \times 0.6838 = 0.561$

Convert to dBi: $G_{\text{achieved,dBi}} = 10 \log(0.561) \approx -2.5$ dBi

Correct choice: (A) -1.32 dBi.